

Updated September 16, 2026

Homework problems for AMAT 540A (Topology I), Fall 2026. Over the course of the semester I'll add problems to this list, with each problem's due date specified. Each problem is worth 2 points.

Solutions will be gradually added (and may be hastily written without proofreading).

Problem 1 (due Weds 9/2): Let $f: A \rightarrow B$ be a function. Suppose that for all $A_0 \subseteq A$, $f^{-1}(f(A_0)) = A_0$. Prove that f is injective.

Solution: Let $a, a' \in A$ with $f(a) = f(a')$. Set $A_0 = \{a\}$, so $a' \in f^{-1}(f(A_0))$. By hypothesis $a' \in A_0$, so $a = a'$. $\square$

Problem 2 (due Weds 9/2): Let A be a set and let $p_1: A \times A \rightarrow A$ be the projection function $p_1(a, a') = a$. Let $R \subseteq A \times A$ be a relation such that the image $p_1(R)$ equals all of A . Prove that if R is symmetric and transitive then it is reflexive.

Solution: Suppose R is symmetric and transitive. Let $a \in A$. Since $p_1(R) = A$ there exists $(a_1, a_2) \in R$ with $p_1(a_1, a_2) = a$. By construction of p this means $a_1 = a$, so a is related to a_2 under R . By symmetry a_2 is related to a , and by transitivity a is related to a , so we conclude R is reflexive. $\square$

Problem 3 (due Weds 9/2): Let $\mathcal{A}$ be a collection of sets and let $U = \bigcup_{A \in \mathcal{A}} A$. Prove that if $\mathcal{P}(U) = \bigcup_{A \in \mathcal{A}} \mathcal{P}(A)$ then $U \in \mathcal{A}$.

Solution: Since $U \in \mathcal{P}(U)$, our hypothesis ensures that $U \in \bigcup_{A \in \mathcal{A}} \mathcal{P}(A)$, i.e., $U \in \mathcal{P}(A)$ for some $A \in \mathcal{A}$. This means $U \subseteq A$, and $A \subseteq U$ by definition, so we conclude $U = A \in \mathcal{A}$ as desired. $\square$

No class Monday 9/7 (Labor Day)

Problem 4 (due Weds 9/9): Prove that $\{U \subseteq \mathbb{N} \mid U = \emptyset \text{ or } |\mathbb{N} \setminus U| < 2026\}$ is not a topology on $\mathbb{N}$.

Solution: Let $U = \mathbb{N} \setminus \{1, \dots, 2025\}$ and $V = \mathbb{N} \setminus \{2026\}$, so U, V are both in this set, but $U \cap V = \mathbb{N} \setminus \{1, \dots, 2026\}$ is not in this set. $\square$

Problem 5 (due Weds 9/9): Let X be a set and $\sim$ an equivalence relation. Write $[x]$ for the equivalence class of $x \in X$ with respect to $\sim$. Prove that $\mathcal{B} = \{[x] \mid x \in X\}$ is a basis for a topology on X .

Solution: Since $x \in [x]$ for all x we have the first axiom. Now suppose $x \in [y] \cap [z]$. Then $x \sim y$ and $x \sim z$, so $[y] \cap [z] = [x] \cap [x] = [x]$, so $x \in [x] \subseteq [y] \cap [z]$ reveals the second axiom. $\square$

Problem 6 (due Weds 9/9): Let $\mathcal{T} = \{U \subseteq \mathbb{R} \mid U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is countable}\}$. Is $\mathcal{T}$ a topology on $\mathbb{R}$? Prove or disprove.

Solution: Yes. First note that $\emptyset \in \mathcal{T}$ by construction, and $\mathbb{R} \in \mathcal{T}$ since $\mathbb{R} \setminus \mathbb{R} = \emptyset$ is countable. Next for any family $U_\alpha \in \mathcal{T}$, without loss of generality none empty, we have that $\mathbb{R} \setminus U_\alpha$ is countable for all α , so $\mathbb{R} \setminus (\bigcup_\alpha U_\alpha)$ is countable (being a subset of any of these), hence $\bigcup_\alpha U_\alpha \in \mathcal{T}$. Finally, for $U, V \in \mathcal{T}$, if one of them is empty their intersection is empty hence in $\mathcal{T}$, and otherwise $\mathbb{R} \setminus U$ and $\mathbb{R} \setminus V$ are countable, hence $\mathbb{R} \setminus (U \cap V) = (\mathbb{R} \setminus U) \cup (\mathbb{R} \setminus V)$ is countable, so $U \cap V \in \mathcal{T}$. $\square$

Problem 7 (due Weds 9/16): Let X be an infinite set with the finite complement topology. Prove that $\mathcal{B} := \{B \subseteq X \mid |X \setminus B| \text{ is a prime number}\}$ is a basis for this topology.

Problem 8 (due Weds 9/16): Let X be a set with at least two elements, with a total order $\leq$ such that for all $a < b$ in X there exists $c \in X$ with $a < c < b$. (Here $<$ means “ $\leq$ and $\neq$ ”.) Prove that the order topology on X is not the same as the discrete topology.

Problem 9 (due Weds 9/16): Let X be the set from Problem 8, with total order $\leq_X$ having the same hypotheses (so the order topology is not discrete). Let Y be another totally ordered set with order $\leq_Y$ and at least two elements. Let $\leq_{lex}$ be the lexicographic order on $X \times Y$. Prove that the order topology on $X \times Y$ using $\leq_{lex}$ does not equal the product topology using $\leq_X$ and $\leq_Y$.

Exam in class on Weds 9/23

Problem 10 (due Weds 9/23): Let X and Y be topological spaces and consider $X \times Y$ with the product topology. Prove that if $C \subseteq X$ and $D \subseteq Y$ are closed then $C \times D$ is closed.

Problem 11 (due Weds 9/23): Let X and Y be infinite sets with the finite complement topology. Prove that the product topology on $X \times Y$ and the finite complement topology on $X \times Y$ are not refinements of each other (note that there are two directions to prove here).

Problem 12 (due Weds 9/23):

Let X be a set with the *excluded point topology* $\{U \mid \spadesuit \notin U\} \cup \{X\}$, where $\spadesuit$ is some fixed element. (It's pretty obvious this is a topology, you don't need to bother proving that.) Prove that X has exactly one set that is both open and dense. [Careful: There's an annoying special case to deal with separately.]

Problem 13 (due Weds 9/30):

Problem 14 (due Weds 9/30):

Problem 15 (due Weds 9/30):
